James Anto

Principal Data Scientist (IC6) / Applied ML, Socure

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Summary

Principal Applied ML / Data Scientist who solves high-stakes fraud and identity decisioning problems. Ships tabular transformers with production explainability, document-intelligence models for regulated risk products, and custom architectures when off-the-shelf fails. Full ML lifecycle ownership plus platform foundations still in production. Hands-on IC technical leader (not a people manager) who raises modeling and hiring bars. Currently enabling governed agent-assisted DS workflows while staying in the loop on modeling judgment.

Experience

Principal Data Scientist, DS Agentic Enablement, Socure – Chennai, India May 2026 – present

- DS Agentic Flywheel: building governed, reviewable agent-assisted workflows so DS teams run real experiments and analysis faster, not demos; still hands-on on fraud and DocV when modeling judgment or evaluation matters.
- DocV Omega developed and shipped in this recent chapter (earlier Predictive DocV work was brief): owned the explainable feature foundation (~2500 unstable candidates to ~200), plus train, eval, reporting, and XAI; guided DS peers in a tight modeling lane.
- Company product context (team/program outcomes, not solo attribution): DocV Omega reports 98% fraud capture in the riskiest 5% with <1% false reject. DocV Omega
- Interview for DS and engineering roles and help keep hiring bars consistent (50+ eng/DS interviews).

Principal Data Scientist, Socure – Chennai, India Jan 2025 – May 2026

- Fraud decisioning needed a path past tree baselines: led early work on Socure's first Sigma-line tabular transformer models (Sigma V4.5 / Fraud V5), owning the modeling approach and the production system around it.
- Owned prep through train, eval, reporting, and explainability; built the explainability path from scratch (8/8 attribution quality checks; ~54 ms P50 / ~275 ms P99) so transformer scores were usable by analysts and downstream systems.
- Technical lead for a small experimentation group (~5) until the path was production-stable; other teams joined later. Company context: Sigma V4.5 tech blog.
- On a ~422K-transaction evaluation, the tabular transformer path outperformed production Sigma V4 by roughly +4% relative PR-AUC and about +2.3 percentage points fraud capture at 5% risk depth.

Data Scientist to Lead Data Scientist, Document Verification, Socure – Chennai, India Mar 2021 – Dec 2024

- Document blur detection: rebuilt the quality model so capture failures stopped breaking downstream DocV (F1 ~0.69 to ~0.96) at very low mobile latency.
- DocV data modeling foundation: ClickHouse + Superset for exploration, plus training and serving plumbing so CV/DL work could iterate against production-scale document data.
- Document understanding: Ravana, a ~2.8M-parameter multilingual KVP transformer built from scratch (Unicode-byte + layout) when off-the-shelf extractors did not fit; brief LatAm and Chinese production before VLMs changed the path.
- Brief Predictive DocV path work (embeddings, watchlist name-match, early supervised DocV fraud framing) that later informed fraud modeling; productized DocV Omega came later.

Senior Software Engineer to Staff Software Engineer (ID+), Socure – Chennai, India Feb 2018 – Feb 2021

- Identity decisioning at production scale: Scala microservices for matching, vendor integrations, and decision/reason-code plumbing so ID+ risk decisions stayed configurable, auditable, and fast enough for customer traffic.
- Platform hardening on the path customers hit every day: observability, ETL/auditing redesigns, transaction search, and security/SSO so failures and audits were operable, not tribal knowledge.

Software Engineer to Senior Software Engineer, Aequalis Software Solutions Pvt. Ltd. – Chennai, India Feb 2014 – Jan 2018

- Employed by Aequalis; day-to-day work on/for Socure in a small founding-era group shipping identity authenticity before a large eng org existed.
- Built early AuthScore product surface and Scala platform foundations, including the EmailAuthScore API still behind Socure's public ID+ evaluate path, plus shared decisioning and rule-code libraries that outlived the original team shape.

Java Developer, Boscosoft Technologies Pvt. Ltd. – India Nov 2011 – Jan 2014
• HR software and a workflow engine (Spring + jsPlumb).

Patents

Generative AI KVP extraction from documents June 2025
James Anto, et al. (Socure)
patents.google.com/patent/US20250182513A1 (USPTO US20250182513A1 (application))

AI and vector-based identity resolution Aug 2025
James Anto, et al. (Socure)
patents.google.com/patent/US20250259177A1 (USPTO US20250259177A1 (application))

KVP extraction using synthetic training data May 2026
James Anto, et al. (Socure)
patents.google.com/patent/US20260134337A1 (USPTO US20260134337A1 (application))

Additional USPTO applications in progress (fraud tokenization; template-based DocV features)
James Anto
((not yet published))

Education

Indira Gandhi National Open University (IGNOU), MCA in Computer Applications – India 2011 – 2013

Indira Gandhi National Open University (IGNOU), BCA in Computer Applications – India 2008 – 2011

Bosco Institute of Information Technology (BIIT), PGDIT in Information Technology – India 2008 – 2011

Skills

Fraud / identity: Tabular transformers, fraud ML, identity verification, feature governance, explainable AI (IG / SHAP)

Applied ML: PyTorch, document AI/CV, custom DL architectures, evaluation and reporting

ML systems: SageMaker, MLflow, Hydra/Lightning, Scala/Python production services

Agentic: Governed agent-assisted DS workflows

Languages: Python (primary), Scala, Java, SQL